Résumé–Job Matching and Talent-Marketplace Engine

Team ID: 8 | Section: 10 | Member: Surabhi Anushka – 2520090168 | Guide: Ms. K. Chandusha
Data Structures and Algorithms – 3 (25CS2103E) | KLH Bachupally Campus, CSE | PBL 2026-27

Abstract

- Data-driven engine that indexes résumés and job descriptions and ranks the best-fit candidates for each role.
- Inverted indexes, priority queues, graph matching and set cover give fast, explainable results.

Problem Statement & Objectives

- **Problem:** Manual screening is slow, biased and misses equivalent skills.
- **Objective 1:** Index and normalize skills.
- **Objective 2:** Explainable candidate ranking.
- **Objective 3:** Optimal role and team assignment.
- **Scope:** Recruiters; matching, ranking, teams.

Existing vs Proposed System

Feature	Existing System	Proposed System
Screening	Manual, slow, labour-intensive	Automated indexing and ranking
Matching	Keyword search; misses “JS” vs “JavaScript”	Skill taxonomy + hash-map normalization
Ranking	Subjective, inconsistent	Live priority-queue ranking
Allocation	Double allocation, manual teams	Graph matching + minimum set cover
Gap	Biased, opaque, hard to scale	Explainable, consistent, scalable

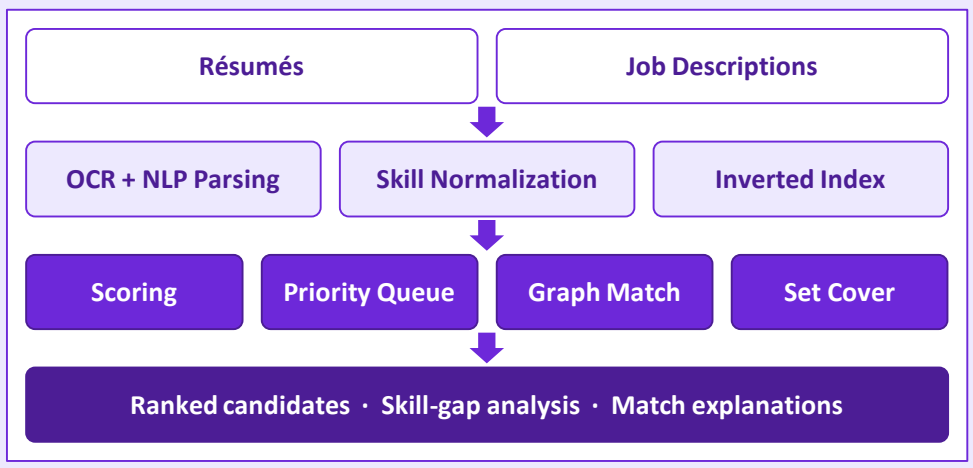
Feasibility Study

- **Technical:** Python with SpaCy/NLTK and custom data structures; runs on an 8 GB RAM laptop.
- **Economic:** Fully open-source stack; no licence cost, cloud hosting optional.
- **Operational:** Simple ranked output with match explanations for recruiters.
- **Schedule:** Extraction, compatibility analysis and applications built in phases within the semester.

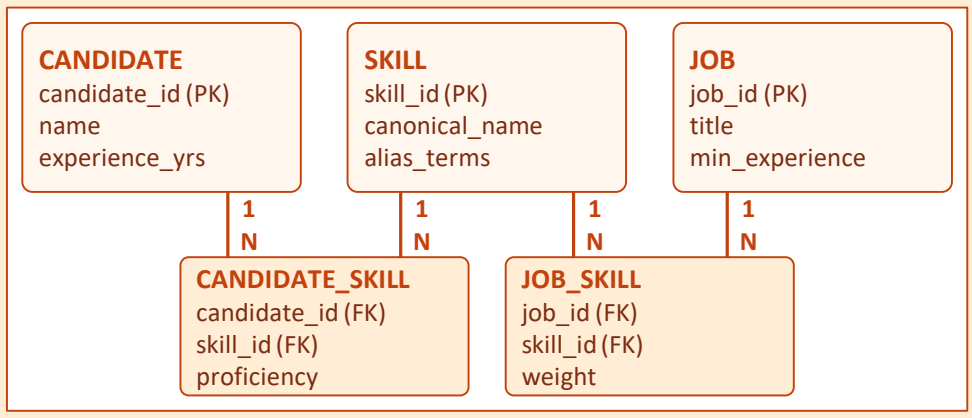
Tools & Technologies

- **Language:** Python 3.10+ (engine logic)
- **NLP:** SpaCy, NLTK (tokenization, lemmatization)
- **Data Structures:** hash maps, inverted indexes, priority queues
- **Database:** SQLite / MongoDB (indexed profiles)
- **Tools:** Git & GitHub, VS Code / PyCharm

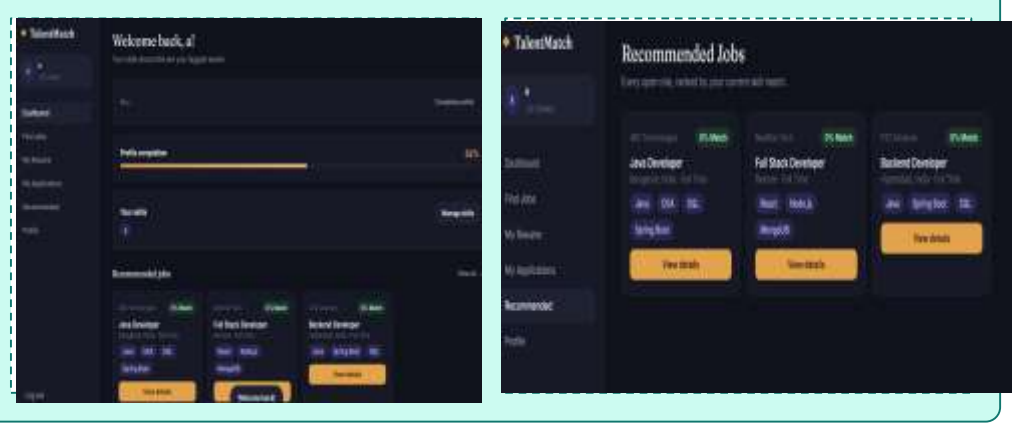
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Results & Testing

- ~72% faster screening
- 100% consistent scoring
- Optimal skill coverage

Conclusion

- Fast, explainable matching built on core data structures.
- Meets all objectives: normalize, rank, assign.

Future Scope

- Predictive success profiles
- Blind-screening bias mitigation
- Multilingual résumé support